

Mathematical Foundations of AI (July/26-Dec/26)

Problem Sheet - 1

Note : Try to solve the following problems through your own efforts without relying on help from external sources like google, AI tools, etc.

1. Consider the set $R1 = \mathcal{R} \setminus \{-1\}$ and define a binary operation $*$: $R1 \rightarrow R1$ defined by $a * b = ab + a + b$ for all $a, b \in R1$.

(a) Show that $(R1, *)$ is an abelian group.

(b) Solve for x satisfying $3 * x * x = 15$ in the abelian group $(R1, *)$.

2. Consider the set $\mathcal{G}$ of 3×3 matrices $\left\{ \begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} : x, y, z \in \mathcal{R} \right\}$. We

define $\cdot$ as the standard matrix multiplication. Is $(\mathcal{G}, \cdot)$ a group ? Is it an abelian group ? Justify your answers.

3. Use Gaussian Elimination to find the set S of all solutions for each of the following systems of linear equations $Ax = b$ where :

(a) $A = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 2 & 5 & -7 & -5 \\ 2 & -1 & 1 & 3 \\ 5 & 2 & -4 & 2 \end{bmatrix} ; b = \begin{bmatrix} 1 \\ -2 \\ 4 \\ 6 \end{bmatrix}.$

(b) $A = \begin{bmatrix} 1 & -1 & 0 & 0 & 1 \\ 1 & 1 & 0 & -3 & 0 \\ 2 & -1 & 0 & 1 & -1 \\ -1 & 2 & 0 & -2 & -1 \end{bmatrix} ; b = \begin{bmatrix} 3 \\ 6 \\ 5 \\ -1 \end{bmatrix}.$

(b) $A = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} ; b = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$

4. Determine the inverse (if it exists) for each of the following matrices :

(a) $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix} ;$ (b) $B = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} ;$

- (c) Define A_x to be the matrix obtained by replacing 6 by $x \in \mathcal{R}$. Determine all values of x for which A_x is invertible.
- (d) Define A_y to be the matrix obtained by replacing $a_{13} = 4$ by $y \in \mathcal{R}$. Determine all values of y for which A_y is invertible.
5. Which of the following are subspaces of $\mathcal{R}^3$?
- (a) $A = \{(\lambda, \lambda + \mu^3, \lambda - \mu^3) : \lambda, \mu \in \mathcal{R}\}$.
 - (b) $A = \{(\lambda^2, -\lambda^2, 0) : \lambda \in \mathcal{R}\}$.
 - (c) Fix any $c \in \mathcal{R}$. Define $A = \{(x, y, z) \in \mathcal{R}^3; x - 2y - 3z = c\}$.
 - (d) $A = \{(x, y, z) \in \mathcal{R}^3 : z \text{ is an integer} \}$.
6. Consider two subspaces of U_1, U_2 of $\mathcal{R}^3$ where U_1 is the solution space of $A_1x = 0$ and U_2 is the solution space of $A_2x = 0$, where
- $$A_1 = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -2 & -1 \\ 2 & 1 & 3 \\ 1 & 0 & 1 \end{bmatrix}; \quad A_2 = \begin{bmatrix} 3 & -3 & 0 \\ 1 & 2 & 3 \\ 7 & -5 & 2 \\ 3 & -1 & 2 \end{bmatrix}.$$
- (a) Determine the dimensions of U_1 and U_2 .
 - (b) Determine a basis for each of U_1 and U_2 .
 - (c) Determine a basis for $U_1 \cap U_2$.
7. Let matrices A_1 and A_2 be as before. Let W_1 and W_2 denote respectively the subspaces spanned by the columns of A_1 and A_2 .
- (a) Determine the dimensions of W_1 and W_2 .
 - (b) Determine a basis for each of W_1 and W_2 .
 - (c) Determine a basis for $W_1 \cap W_2$.
8. Let A and b be the same as defined in Question 3(a). For each of the following modifications of A 's entries, find out the ranges of x for which the system is solvable or not solvable :
- (a) $a_{22} = 2$ is replaced by x .
 - (b) $a_{24} = -5$ is replaced by x .

9. For $\lambda \in \mathcal{R}$, define $A_\lambda = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & -7 \\ 2 & -1 & \lambda \end{bmatrix}$. Apply Gaussian-Elimination to
- Determine all values of $\lambda \in \mathcal{R}$ for which A_λ is invertible.
 - For each λ for which A_λ is invertible, determine the inverse A_λ^{-1} .
 - For each λ for which A_λ is not invertible, determine a $x \in \mathcal{R}^3$ satisfying $x \neq 0$ and $A_\lambda x = 0$.
 - For each λ for which A_λ is not invertible, describe mathematically the Null-space and range space of A_λ and also determine their dimensions.
10. Prove the following : If U and W are subspaces of a finite-dimensional vector space V such that $U \cap W = \{0\}$, then any $u \in U$ is independent of any collection of vectors from W and any $w \in W$ is independent of any collection of vectors from U .
11. Define $b_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, $b_2 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$, $b_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$. Define $C = (e_1, e_2, e_3)$ to be the ordered standard basis of $\mathcal{R}^3$.
- Prove that $B = (b_1, b_2, b_3)$ is an ordered basis of $\mathcal{R}^3$.
 - Let $T : \mathcal{R}^3 \rightarrow \mathcal{R}^3$ be the linear transformation represented by
$$A \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} 3x_1 + 2x_2 + x_3 \\ x_1 + x_2 + x_3 \\ 2x_1 + 3x_2 + x_3 \end{bmatrix}$$
with respect to standard basis C for both the domain and co-domain. Determine A and also the matrix representation of T with respect to basis B for both the domain and co-domain.
12. For $\mu \in \mathcal{R}$ satisfying $\mu^2 \leq 1$, define $x(\mu)$ to be the 3-vector $(\frac{\sqrt{3}\mu}{2}, \frac{\mu}{2}, \sqrt{1-\mu^2})$. Define $U := \text{span}(\{x(\mu) : \mu^2 \leq 1\})$. Provide a brief justification for each of your answers.

- (a) Determine an ordered basis B and also the dimension d of U .
 - (b) Express an arbitrary element $(x, y, z) \in U$ as a linear combination of elements of B .
 - (c) Determine a subspace W of $\mathcal{R}^3$ of dimension $3 - d$ for which $\mathcal{R}^3 = U \oplus W$. $\oplus$ represents the direct sum of U and W .
13. Given the basis $C = \{c_1, c_2, c_3\}$ where $c_1 = (2, 0, 0)$, $c_2 = (1/\sqrt{2}, 1/\sqrt{2}, 0)$, $c_3 = (3/8, 3/8, 1/4)$ of R^3 , apply Gramm-Schmidt Algorithm to find an orthonormal basis for R^3 .
14. Given the standard basis $B = \{e_1, e_2, e_3\}$ of R^3 and also the basis C (stated in the previous problem), construct the matrices P and Q representing the transformations mapping B to C and C to B respectively.